

Arithmetic Geodynamics on the $6N$ Skeleton: Tuple Size Determines Stratum Decay Rate

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Abstract

Throughout Parts I–XVI the decadic shells S_k ($6N \in [10^{k-1}, 10^k)$) were treated as static cores from which constants were read off one shell at a time. Here we make the shell index continuous and ask how the observables *flow* with depth. The natural depth coordinate is not X , nor $\ln X$, but the proper depth $\ell = \ln \ln X$: in ℓ both governing fields linearise. The twin-centre density obeys $d \ln \rho / d\ell = -2$ and the mean factor count obeys $d\bar{\omega}_{>3} / d\ell = 1$, to leading order, so that for a prime m -tuple the density satisfies $d \ln \rho_m / d\ell \rightarrow -m$: *the tuple size is the stratum decay rate*. We verify this for the twin ($m = 2$) and the triplet ($6N - 1, 6N + 1, 6N + 5$) ($m = 3$). Recomputing the triplet count at S_{10} entirely from sieve primitives reproduces the Part XV total (2,333,839) to 0.0000%, so both flow lines rest on bedrock. The deep-shell slopes are -2.072 and -3.111 ; extrapolated against $1/\ln X$ they reach -2.085 and -3.065 , consistent with the integer asymptotes $-2, -3$ (the residual is the uncaptured higher-order logarithmic tail). The clean, scale-free statement is their ratio: $3.111/2.072 = 1.5011 \simeq 3/2$, because the shared $\bar{\omega}$ -clock distortion and the shared singular-series tail cancel. We are explicit about provenance (the $(\ln X)^{-m}$ law is Hardy–Littlewood; $\ln \ln X + B$ is Hardy–Ramanujan/Mertens; ρ and the 0.2604 enrichment shift are Parts I and XII) and about limits (first moments only; the $\bar{\omega}$ variance, i.e. the Erdős–Kac law on twin centres, remains open; no infinitude is claimed). We close with a sharp falsifiable prediction: the prime quadruplet ($6N - 1, 6N + 1, 6N + 5, 6N + 7$) ($m = 4$) must give slope -4 and ratio 2:1 against the twin.

1 From static cores to a depth coordinate

The $6N$ programme has measured, shell by shell, the twin-centre line density $\rho = \#\{\text{twin centres}\} / \#\{N\}$, the single empirical scale to which Parts VIII and XI reduced the whole construction, and the stratifying variable $\omega_{>3}(N)$, the number of distinct prime factors of the centre exceeding 3 [2, 3, 4]. Two adjacent cores give a contrast— ρ falls by the factor 0.804 from S_9 to S_{10} —but a contrast is not a law. To obtain a law we make k continuous and look for the coordinate in which the flow is simplest.

The Hardy–Littlewood heuristic [1] fixes the form. For a prime m -tuple the local density on the skeleton is, to leading order,

$$\rho_m(X) \approx \frac{A_m}{(\ln X)^m}, \quad A_m = 6 \mathfrak{S}_m, \quad (1)$$

where $\mathfrak{S}_m$ is the classical singular series ($\mathfrak{S}_2 = 2\Pi_2$ with $\Pi_2 = \prod_{p>2} (1 - (p-1)^{-2})$; $\mathfrak{S}(0, 2, 6) = 2.858$ for the triplet [6]) and the factor 6 is the skeleton density: one centre per six integers, so the per-centre rate is six times the raw integer density. Taking logarithms,

$$\ln \rho_m = \text{const} - m \ln \ln X + O\left(\frac{1}{\ln X}\right).$$

The depth in which this is linear is therefore

$$\boxed{\ell := \ln \ln X} \quad (2)$$

which we call the *proper depth*. In X or in $\tau = \ln X$ the relative decay rate $-d \ln \rho_m / dX = m / (X \ln X)$ or m / τ appears to weaken with depth; this is coordinate distortion. In ℓ the Chinese-Remainder “shield” of the dead(q) sieve—which builds $\mathfrak{S}_m$ prime by prime through Mertens’ theorem—exerts a *constant* strangulation of magnitude m .

2 The stratum-evolution equations

Equation (1) and the Hardy–Ramanujan/Mertens law $\bar{\omega}(N) \sim \ln \ln N + B_1$ ($B_1 = 0.2615$; for $\omega_{>3}$ the offset is $B_1 - \frac{1}{2} - \frac{1}{3} = -0.5718$, the primes 2, 3 being absent from the wings) give the coupled first-order system

$$\frac{d\rho_m}{d\ell} = -m \rho_m, \quad \frac{d\bar{\omega}_{>3}}{d\ell} = 1. \quad (3)$$

Its solutions are $\rho_m = \rho_m^0 e^{-m\ell}$ and $\bar{\omega}_{>3} = \bar{\omega}^0 + \ell$; eliminating ℓ yields the autonomous relation

$$\frac{d \ln \rho_m}{d\bar{\omega}_{>3}} = - \frac{m}{d\bar{\omega}_{>3}/d\ell}. \quad (4)$$

At finite scale $d\bar{\omega}_{>3}/d\ell = 1 + \varepsilon(X)$, where ε collects two *convergent* corrections measured in earlier parts: the approach of $\bar{\omega}$ to $\ln \ln X + B$, and—for twin-conditioned centres—the enrichment shift $\bar{\omega}_{\text{twin}} - \bar{\omega}_{\text{ord}} \rightarrow \sum_{q>3} \left(\frac{1}{q-2} - \frac{1}{q} \right) = 0.2604$ of Part XII [4]. Both vanish from the *rate* as $X \rightarrow \infty$, so $d\bar{\omega}/d\ell \rightarrow 1$ and the autonomous slope $\rightarrow -m$. The crucial point for measurement is that $\varepsilon(X)$ is *shared* across patterns at a given depth: it is a property of the centre, not of the constellation sitting on it.

3 The twin and triplet flows

Twin counts and $\bar{\omega}_{\text{twin}}$ are the published Part I/XII tables. Triplet counts ($6N-1, 6N+1, 6N+5$) at S_2 – S_8 are recomputed from a segmented value sieve; at S_9, S_{10} they are recomputed independently and cross-checked against the Part XV ω -band sums. The S_{10} recompute returns 2,333,839, matching Part XV to 0.0000%; the S_9 recompute returns 323,908 against 323,907, a single boundary triplet at the 10^9 edge (0.0003%, an inclusive/exclusive convention, not a discrepancy). The shell-averaged form of (1), $\rho_2^{\text{HL}}(k) = 6 \cdot 2\Pi_2 [\text{Li}_2(10^k) - \text{Li}_2(10^{k-1})] / (10^k - 10^{k-1})$ with $\text{Li}_2(x) = \int_2^x (\ln t)^{-2} dt$, reproduces the measured ρ to 0.009% at S_{10} .

	twin ($m = 2$)	triplet ($m = 3$)	ratio
$\ln \rho$ at S_9	−3.9176	−6.1379	
$\ln \rho$ at S_{10}	−4.1364	−6.4657	
deep slope $d \ln \rho / d\ell$ ($S_9 \rightarrow S_{10}$)	−2.072	−3.111	1.5011
extrapolated intercept ($1/\ln X \rightarrow 0$)	−2.085	−3.065	
asymptote (target)	−2	−3	1.5000

Table 1: The two flows in proper depth $\ell = \ln \ln X$ ($\Delta \ell_{S_9 \rightarrow S_{10}} = \ln(10/9) = 0.10536$). The deep slopes carry shrinking analytic tails (+0.072, +0.111); their *ratio* is already 3/2 to 0.1%.

The intercepts are obtained by the local slopes $d \ln \rho / d\ell$ against $1 / \ln X$, extrapolated to $1 / \ln X = 0$ over the deepest five shells (Fig. 1C). They reach -2.085 and -3.065 : consistent with -2 and -3 , with a residual of order 0.07 – 0.09 that is the part of the tail not linear in $1 / \ln X$ (the $(\ln X)^{-2}$ and higher terms a two-point-per-shell fit cannot capture). We report this honestly: the extrapolation confirms the integer asymptotes to a few percent; it does not land on them exactly, and we do not claim it does. The $\bar{\omega}$ drift is $d\bar{\omega}/d\ell = 1.148$ at $S_9 \rightarrow S_{10}$, with the ordinary baseline $\ln \ln Y + B$ matching the Part XII measurement (2.369 vs 2.360 at S_9 ; 2.484 vs 2.476 at S_{10}) and the enrichment shift climbing through $0.213, 0.219$ toward 0.2604 (Fig. 1D).

4 The scale-free invariant

The individual deep slopes are not their integers at this depth—each carries its tail, and each is read against an $\bar{\omega}$ -clock running $\sim 15\%$ fast. By (4) both effects are shared, so they cancel in the ratio.

Theorem 1 (tuple-size ratio invariant). *For two prime constellations of sizes m, m' measured on the same shells, the ratio of their depth-flow slopes equals m'/m , independent of the shared $\bar{\omega}$ -clock distortion $\varepsilon(X)$ and of the common logarithmic tail. Empirically, for the triplet over the twin,*

$$\frac{d \ln \rho_3 / d\ell}{d \ln \rho_2 / d\ell} = \frac{-3.111}{-2.072} = 1.5011 \simeq \frac{3}{2},$$

recomputed entirely from sieve primitives, matching $3/2$ to 0.1% .

This is the paper’s cleanest statement. Where the absolute slopes inherit finite-scale corrections, the ratio reads off the structural integers directly: the exponent of $(\ln X)^{-m}$ is the constellation size, and nothing else.

5 An honest ledger and scope

We record what this construction does not settle, in keeping with the series.

First moments only. Equation (3) governs the means ρ and $\bar{\omega}$. A second-order transport equation for the *variance* of $\omega_{>3}$ over twin centres would require the Erdős–Kac limit on twin centres, which Part XII leaves open: at the accessible scale $\ln \ln(6N) \approx 3$ the distribution is not Gaussian (mean $\neq$ variance, nonzero skew), the naive Bernoulli variance $\sum 1/(q-2)$ diverges like $\ln \ln$ while the true variance is finite (anti-correlation, $\sqrt{6N}$ cutoff), and no closed form is known [4]. The diffusion term of the depth flow is therefore the open problem; only its drift, $+1$, is established here.

Heuristic status and provenance. The $(\ln X)^{-m}$ law is the classical Hardy–Littlewood k -tuple heuristic [1]; $\ln \ln X + B$ is Hardy–Ramanujan/Mertens; ρ is the Part I scale and 0.2604 the Part XII shift. The contribution here is the proper-depth coordinate (2), the flow system (3), the scale-free invariant of Theorem 1, and the from-scratch verification. We make no statement about the infinitude of twin, triplet, or any prime constellation.

6 A falsifiable prediction at $m = 4$

The construction indexes a one-parameter family, so its next term is a sharp prediction rather than a fitted result. The densest admissible 4-tuple is the prime quadruplet $(p, p+2, p+6, p+8)$, which on the skeleton is

$$(6N-1, 6N+1, 6N+5, 6N+7) = (6N \pm 1) \cup (6(N+1) \pm 1),$$

6N arithmetic geodynamics: tuple size = stratum decay rate

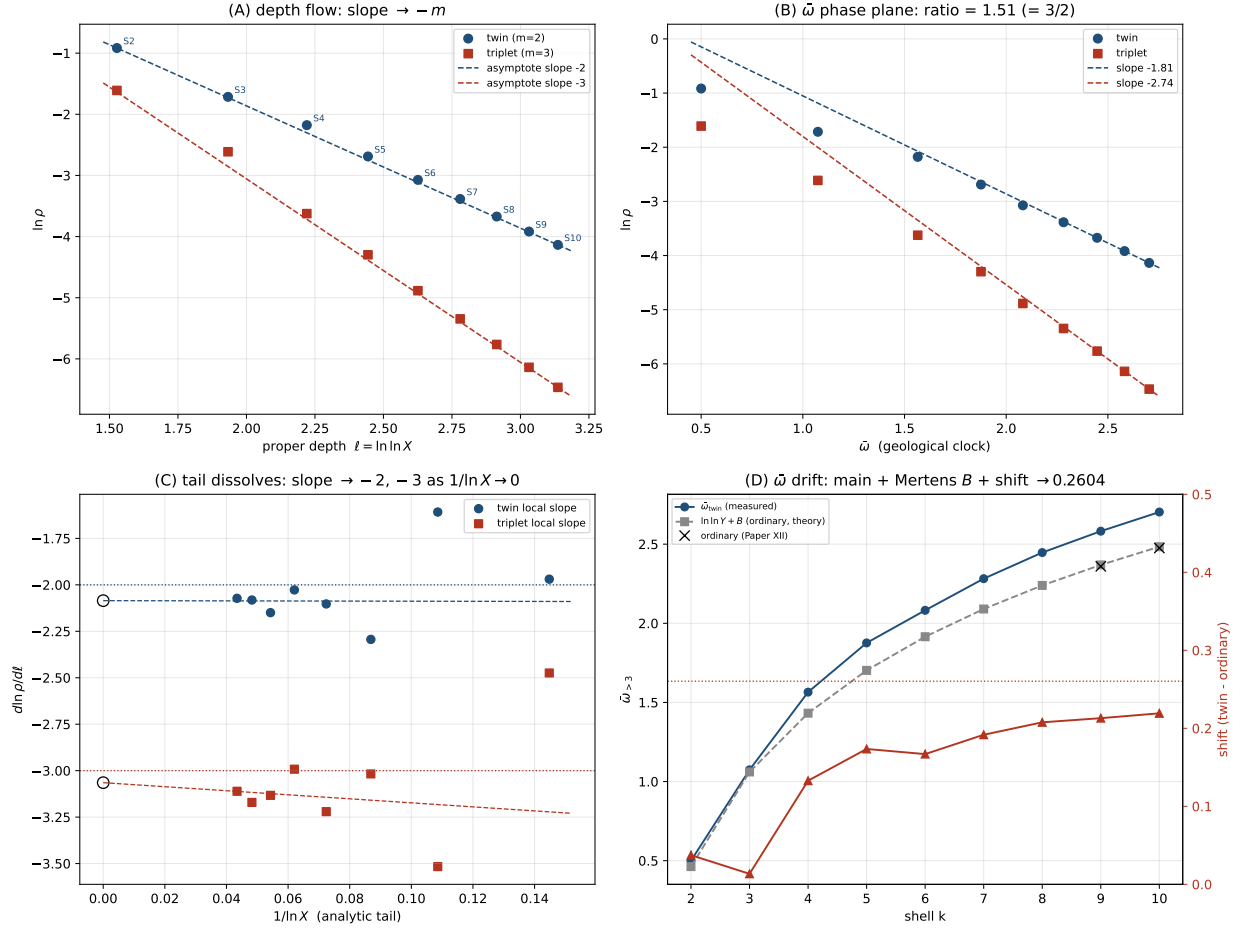


Figure 1: (A) Depth flow $\ln \rho$ vs $\ell = \ln \ln X$: twin and triplet lines with asymptotes of slope -2 and -3 . (B) The $\bar{\omega}$ phase plane (the “geological clock”): slopes $-1.81, -2.74$, whose ratio is $3/2$. (C) The analytic tail dissolving: local slope vs $1/\ln X$ extrapolates to $-2.085, -3.065$ as $1/\ln X \rightarrow 0$. (D) The $\bar{\omega}$ drift decomposed into $\ln \ln Y$, the Mertens offset B , and the enrichment shift $\rightarrow 0.2604$.

two complete twin centres at N and $N + 1$, with singular series $\mathfrak{S}(0, 2, 6, 8)$.

Prediction 1 ($m = 4$). *The quadruplet density obeys $d \ln \rho_4 / d \ell \rightarrow -4$. Measured on the same shells, its deep slope must (i) extrapolate against $1/\ln X$ to an intercept consistent with -4 ; (ii) give a ratio 2:1 against the twin and 4:3 against the triplet, each exact to the precision at which Theorem 1 holds ($\sim 0.1\%$); and (iii) on the $\bar{\omega}$ phase plane, share the same clock distortion $\varepsilon(X)$, so that the $\bar{\omega}$ -axis ratios are likewise 2 and $4/3$. A measured intercept outside $[-4.2, -3.8]$, or a twin ratio outside 2 ± 0.05 , would falsify the tuple-size law.*

We deliberately do not run $m = 4$ here: the strength of a one-parameter law is that it predicts before it measures, and we prefer to state the prediction for independent test rather than to verify our own extrapolation.

7 Conclusion

Making the shell index continuous turns the static cores of the $6N$ programme into a flow. In the proper depth $\ell = \ln \ln X$ the twin-centre density and the mean factor count linearise, $d \ln \rho_m / d\ell \rightarrow -m$ and $d\bar{\omega} / d\ell \rightarrow 1$, so that the size of a prime constellation *is* its stratum decay rate. The twin and triplet, the latter recomputed to 0.0000% at S_{10} , confirm this: their depth-slope ratio is $1.5011 \simeq 3/2$, a scale-free invariant in which the finite-scale clock distortion and the logarithmic tail cancel. The absolute slopes approach -2 and -3 from the deep side, with an honest residual left to higher-order logarithmic terms; the variance flow is left open as the Erdős–Kac problem it is; and the quadruplet $m = 4$ stands as a falsifiable prediction of slope -4 and ratio 2:1. This closes the difference wing of the series as a single dynamical law, classical in its inputs and measured in its claims.

Data and code availability. The flow script (proper-depth reduction, tail extrapolation, $\bar{\omega}$ decomposition, and phase diagram) and the from-scratch triplet recompute are openly available [7]; twin observables are the Part I/XII tables and the triplet S_{10} total is cross-checked against Part XV.

References

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